Proof: If $P \neq L$, Then Some Language in P Is Not L-Complete

Claim. If $P \neq L$, then there exists a language in P that is not L-complete under logspace reductions.

Proof. Assume $P \neq L$. Then, by definition of inequality of complexity classes, there exists some language

$$A \in P \setminus L.$$

So $A \in P$ and $A \notin L$.

We claim that A is not L-complete under logspace reductions.

Recall that a language C is L-complete under logspace reductions only if both of the following hold:

1. $C \in L$;
2. for every language $B \in L$, we have $B \leq_m^{\log} C$.

In particular, every L-complete language must itself belong to L.

But our chosen language A does not belong to L. Therefore A cannot be L-complete under logspace reductions.

Hence there exists a language in P that is not L-complete under logspace reductions. $\square$